DOMAIN 02

# Web Development

Technical Club Recruitment Task: Website Development | Choose tasks by your year

## FIRST-YEAR STUDENTS

**Objective:** Build a creative and visually appealing website using HTML and CSS, with optional JavaScript. This evaluates your creativity, web development fundamentals and problem-solving skills. Choose **ONE** task.

| Task | Requirements |
|---|---|
| 1. Personal Portfolio Website | Create a personal website to introduce yourself and showcase your interests. Sections: Home with your name and introduction, About Me, Skills and interests, Projects or achievements (if any), Contact or social links. |
| 2. Mini Games Website | A simple website with browser-based games: attractive homepage, at least one playable game, clear instructions and result display, a restart button. Suggested: Rock Paper Scissors, Tic-Tac-Toe, Number Guessing Game. |
| 3. Gamified Portfolio Website | Turn a personal portfolio into a game-inspired experience: personal introduction and profile, skills as levels or achievements, a progress bar or achievement badges, at least one interactive feature. |
| 4. Animated Website | Show creativity with CSS animations: attractive hero section, at least two CSS animations, hover effects on buttons or cards, creative typography, colors and layout. |

### TECHNICAL REQUIREMENTS

- Use HTML5 and CSS3.
- JavaScript is optional, except where required for game functionality or interactive features.
- No frameworks or libraries such as React, Bootstrap or Tailwind CSS.
- Focus on clean code, good design and working features.

### LIVE DEMONSTRATION GUIDELINES

- Bring your laptop with your project ready to run.
- Demonstrate your project live and explain its key features.
- Be prepared to explain your code, technology choices and implementation.
- If deployed, you may also showcase the live website.
- Keep your GitHub repository and README ready for reference.

> **SUBMISSION: BRING EVERYTHING ON THE DAY OF AUDITION.** There is **no online submission**: do not send or upload your project anywhere. Bring your laptop with the project ready to run, and keep your GitHub repository, README and deployed link (if any) ready to show.

### IMPORTANT NOTES

- Choose only ONE task.
- You are not expected to know everything beforehand. Focus on learning and implementing the basics.
- Use of AI is allowed, but you must be able to debug and change small functionalities (if asked) during the interview.
- A simple, well-executed project is better than an incomplete, complicated one.
- Deployment is encouraged but not compulsory.

**All the best! Be creative, experiment, and have fun building your first website.**

# SECOND-YEAR STUDENTS

Choose any **ONE** of the three tasks.

### Task 1: Operation Hidden Code (Frontend Development)

**Tech stack:** HTML, CSS, JavaScript

Build a futuristic mystery website where users explore an interactive interface, solve puzzles and uncover a secret code.

**Mandatory:**

- Creative UI with animations and visual effects.
- Interactive terminal with at least five custom commands.
- At least three connected puzzles.
- One hidden Easter egg.
- Final secret-code reveal.
- Feedback for correct and incorrect answers.

**Bonus:** Timer, scoring system, sound effects, saved progress, or multiple endings.
**Bring on audition day:** GitHub repository, deployed website and a short demo (all shown live on your laptop).

### Task 2: Full-Stack Development (Build Your Own Product)

**Tech stack:** Open choice (MERN, Next.js, Django or any suitable stack)

Build a complete, functional web application that solves a real-world problem or provides an original user experience.

**Mandatory:**

- Interactive frontend.
- Backend APIs with meaningful business logic.
- Database integration.
- User authentication and appropriate access control.
- Input validation and error handling.
- At least one feature beyond basic CRUD operations.
- Deployed working application.

**Suggested ideas:**

- Collaborative workspace with real-time updates.
- Gamified coding platform with leaderboards.
- Campus community platform.
- Hackathon teammate-matching platform.
- An original application idea.

**Bonus:** AI integration, real-time communication, automated tests, or an admin dashboard.
**Bring on audition day:** GitHub repository, deployed application, README, and a demo explaining the architecture and technical decisions (all shown live on your laptop).

### Task 3: Innovation Project (Think Beyond the Ordinary)

**Tech stack:** Open choice

Identify a real-world problem and develop an innovative solution using technology. It may involve web development, mobile applications, AI/ML, IoT, automation, cybersecurity or any other relevant field.

**Requirements:**

- Identify a genuine problem and its target users.
- Propose an original or meaningfully improved solution.

- Develop a working prototype or minimum viable product (MVP).
- Demonstrate the core functionality.
- Explain the technology used and why it was chosen.
- Describe the potential impact, limitations and future scope.

**Bonus:** A unique technical approach, user testing, measurable results, or a clear plan for real-world implementation.

## LIVE DEMONSTRATION GUIDELINES

- Bring your laptop with your project ready to run.
- Demonstrate your project live and explain its key features.
- Be prepared to explain your code, technology choices and implementation.
- If deployed, you may also showcase the live website.
- Keep your GitHub repository and README ready for reference.

> **SUBMISSION: BRING EVERYTHING ON THE DAY OF AUDITION.** There is **no online submission**: do not send or upload your project anywhere. Bring your laptop with the project ready to run, and keep your GitHub repository, README and deployed link ready to show.

## GENERAL GUIDELINES

- Choose any ONE of the three tasks.
- Task 1 must use HTML, CSS and JavaScript without frontend frameworks.
- Task 2 and Task 3 allow suitable technologies of your choice.
- Use of AI is allowed, but you must be able to debug and change small functionalities (if asked) during the interview.
- A working, well-explained project is valued more than unnecessary complexity.

**Don't just build a project. Build something worth remembering.**

# THIRD-YEAR STUDENTS: ADVANCED DEVELOPMENT & INNOVATION

Choose **ONE** of the following challenges. Tech stack for all tasks: open choice.

### Task 1: Build a Production-Ready Web Application

Design and develop a full-stack application that solves a real-world problem and demonstrates production-level engineering practices.

**Mandatory requirements:**

- Responsive and user-friendly interface.
- Frontend, backend and database integration.
- Authentication and role-based access control where appropriate.
- Proper API design, validation and error handling.
- Secure handling of user data and sensitive configuration.
- At least one advanced feature, such as real-time updates, background processing, analytics or intelligent recommendations.
- Deployment with a working demonstration.

**Bonus:** Automated tests, CI/CD, caching, logging, monitoring or containerization.

### Task 2: AI-Powered Innovation Challenge

Build an AI-powered application that solves a meaningful problem using AI, machine learning, intelligent automation or data-driven decision-making.

**Mandatory requirements:**

- Clearly defined problem statement and target users.
- A working prototype with a meaningful AI/ML component.
- Integration with an application interface or workflow.
- Evaluation of output quality using suitable test cases or metrics.
- Handling of invalid inputs, failure cases and limitations.
- Explanation of the model, approach and technology choices.

**Bonus:** Retrieval-Augmented Generation (RAG), agentic workflows, local models, model evaluation, or cost and latency optimization.

### Task 3: Open Innovation (Build Something Impactful)

Identify a real-world problem and create an original technological solution. You may work in domains such as healthcare, education, agriculture, cybersecurity, accessibility, sustainability, smart cities or developer tools.

**Mandatory requirements:**

- Identify the problem and intended users.
- Propose a solution with a clear innovation factor.
- Build a working prototype or minimum viable product (MVP).
- Demonstrate the core functionality using a realistic scenario.
- Explain the architecture, technology choices and implementation.
- Discuss feasibility, limitations and future improvements.

**Bonus:** User validation, measurable impact, novel algorithms, hardware integration, or a credible path to real-world adoption.

## LIVE DEMONSTRATION GUIDELINES

- Bring your laptop with the project ready to run.

- Demonstrate the working application or prototype live.
- Explain the architecture, key implementation decisions and technical challenges.
- Be prepared to answer questions about your code and show how your solution handles realistic scenarios.
- Keep your GitHub repository, README and deployment link ready for reference, if applicable.

> **SUBMISSION: BRING EVERYTHING ON THE DAY OF AUDITION.** There is **no online submission**: do not send or upload your project anywhere. Bring your laptop with the project ready to run, and keep your GitHub repository, README, and deployment link (if applicable) ready to show.

## IMPORTANT NOTES

- Choose only ONE task.
- You may use any suitable programming language, framework or technology.
- Use of AI is allowed, but you must be able to debug and change small functionalities (if asked) during the interview.
- A working, well-tested solution is more valuable than a feature-heavy but unreliable prototype.

**The challenge: Don't just demonstrate what you know. Demonstrate what you can engineer.**